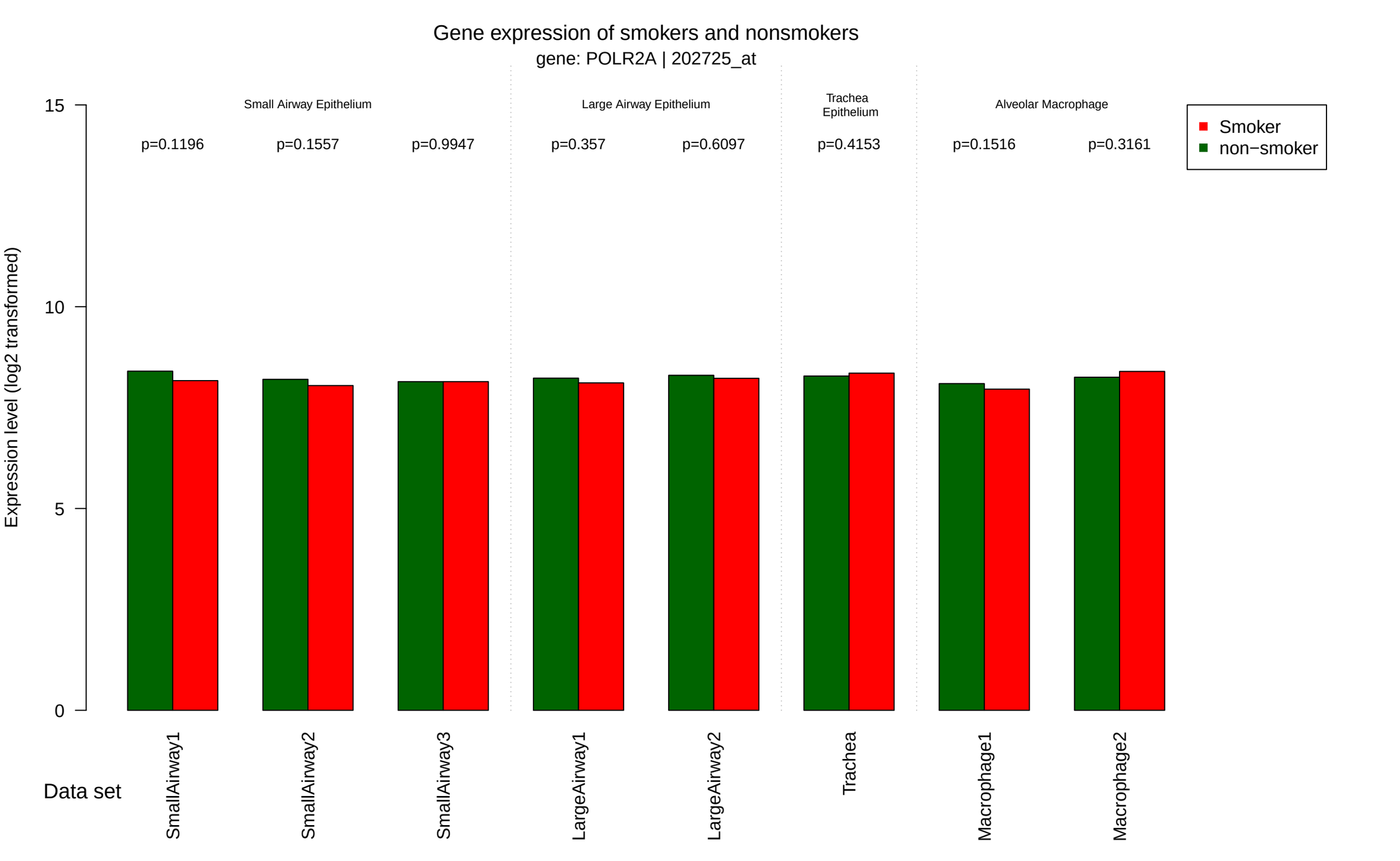
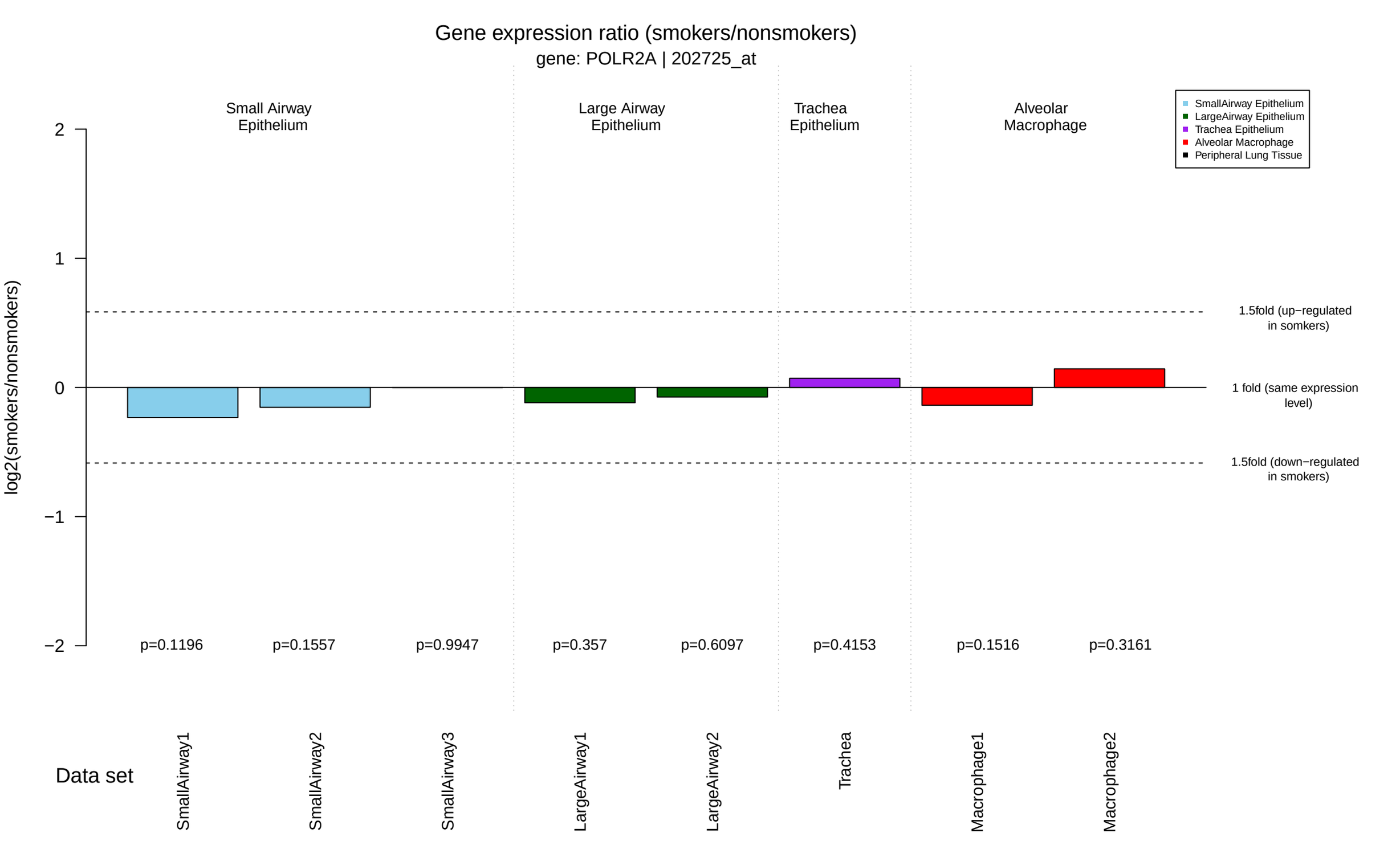
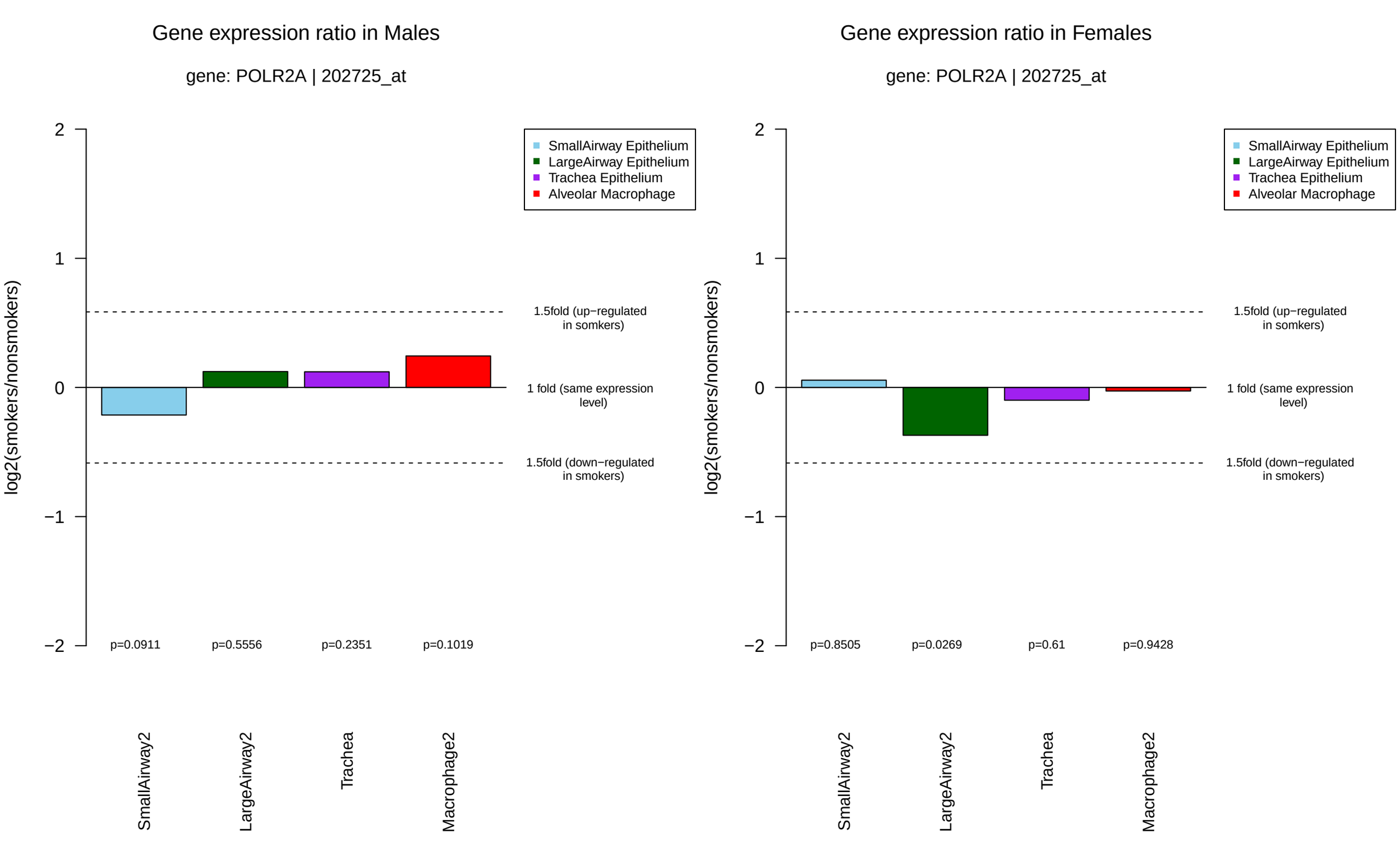
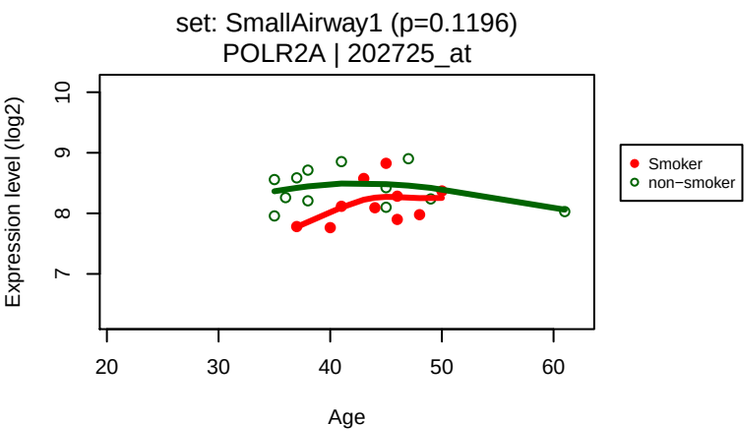
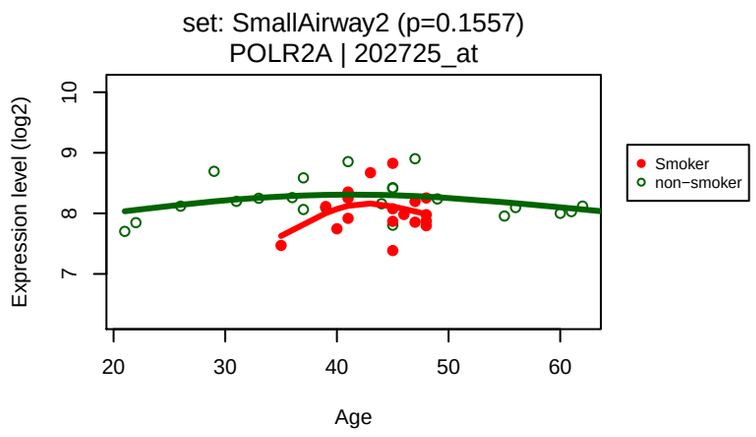
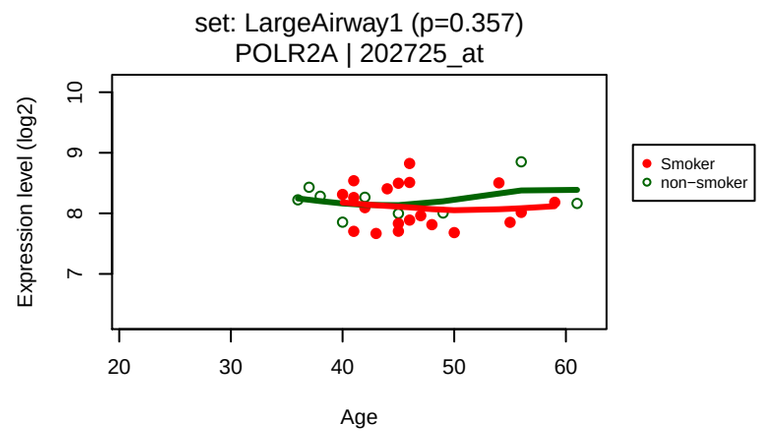

Gene expression of smokers and nonsmokers
gene: POLR2A | 202725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1196
p=0.1557
p=0.9947
p=0.357
p=0.6097
p=0.4153
p=0.1516
p=0.3161
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

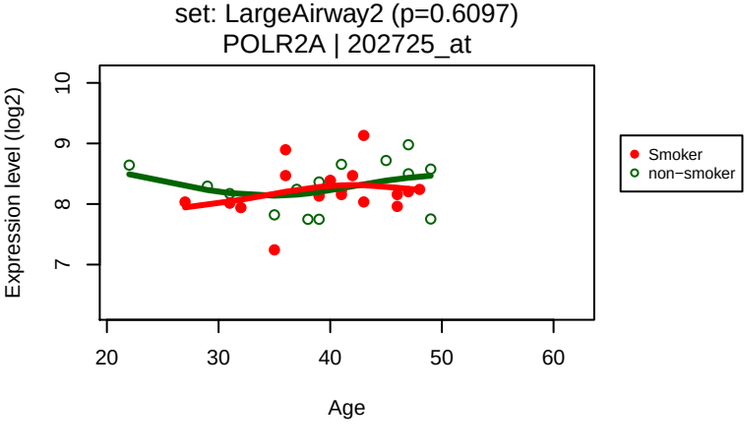

Gene expression ratio (smokers/nonsmokers)
gene: POLR2A | 202725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1196
p=0.1557
p=0.9947
p=0.357
p=0.6097
p=0.4153
p=0.1516
p=0.3161
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

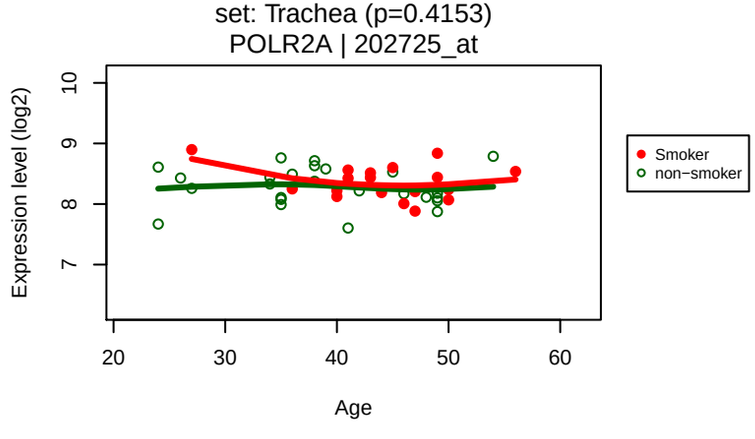

Gene expression ratio in Males
gene: POLR2A | 202725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0911
p=0.5556
p=0.2351
p=0.1019
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLR2A | 202725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8505
p=0.0269
p=0.61
p=0.9428
SmallAirway2
LargeAirway2
Trachea
Macrophage2

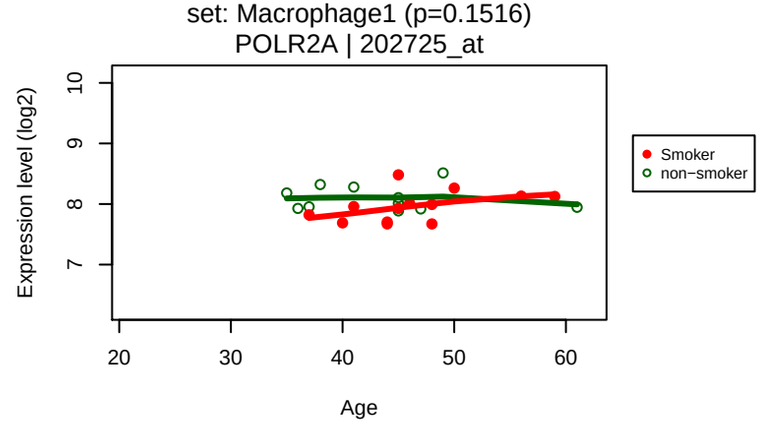

set: SmallAirway1 (p=0.1196)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker

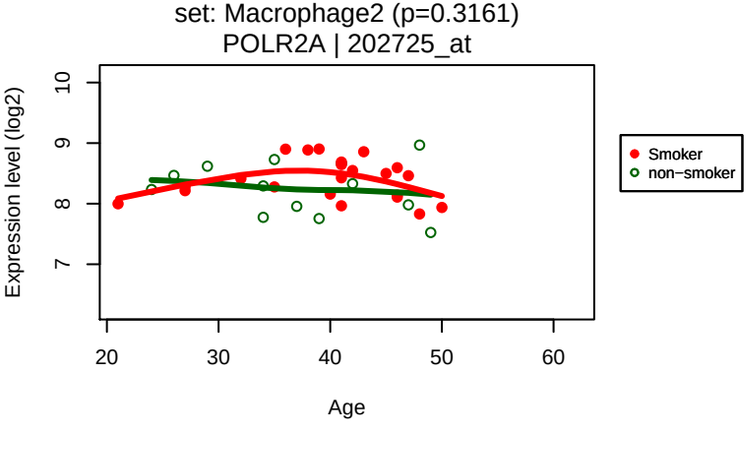

set: SmallAirway2 (p=0.1557)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker

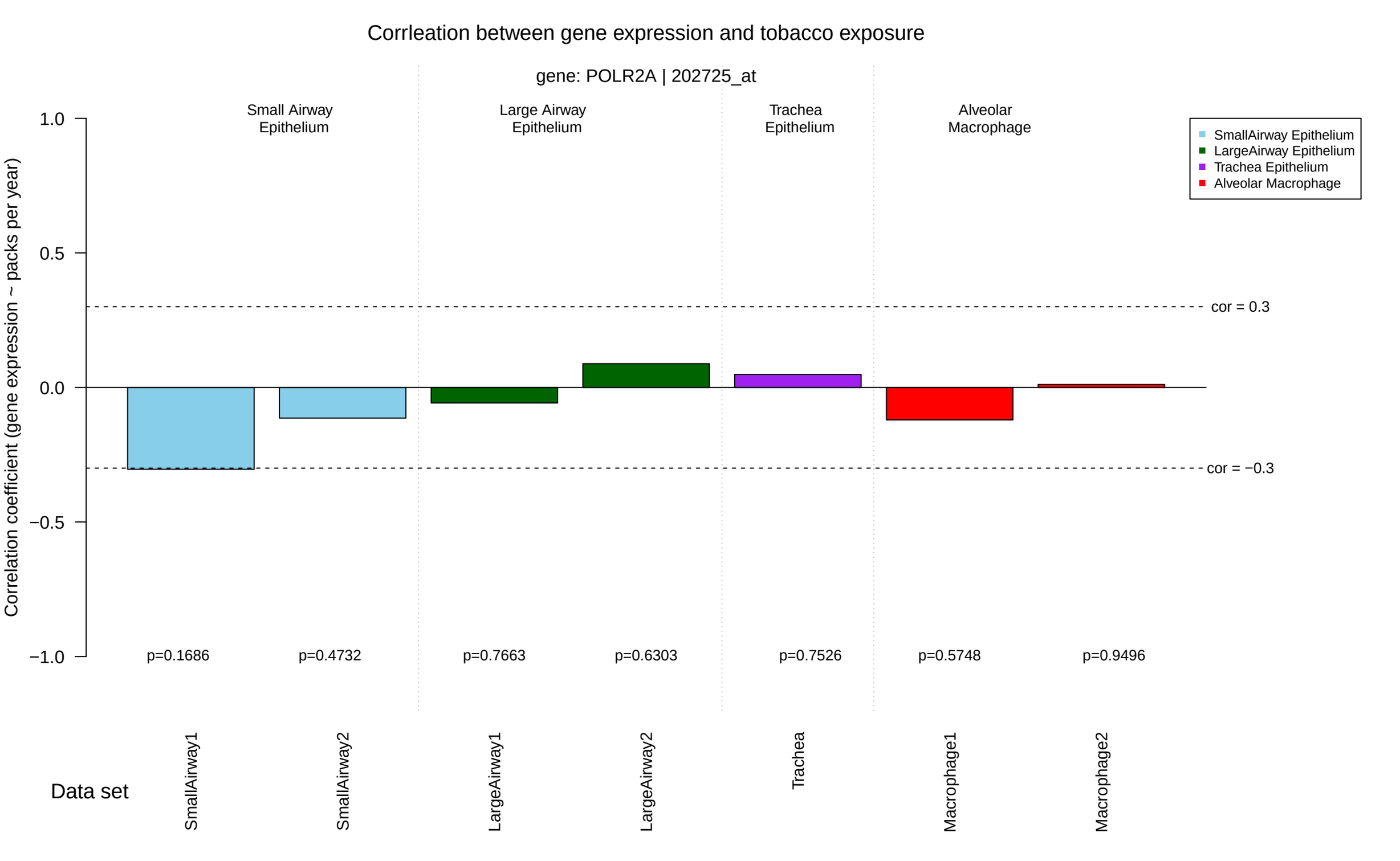

set: LargeAirway1 (p=0.357)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6097)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4153)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1516)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3161)
POLR2A | 202725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLR2A | 202725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1686
p=0.4732
p=0.7663
p=0.6303
p=0.7526
p=0.5748
p=0.9496
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2